THE ALGEBRAIC ELEMENTS OF AN EXTENSION FORM A SUBEXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file fields.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: The separable-side twin of lemma-purely-inseparable-elements (b7 companion card L48372).

Editorial note. The label lemma-algebraic-elements names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- Chapter lemma-algebraic-finitely-generated (source lemma, worklist row 72): a finitely generated algebraic extension is finite; here k(alpha, beta)/k is generated by the two algebraic elements alpha, beta
- Chapter lemma-finite-is-algebraic (delivered card L48360): a finite extension is algebraic; subextensions of finite extensions are finite

Source reference. Original reference 09GH: sources/stacks-fields/pinned-original-source.tex, source lines 910–910. Prerequisite pointer invoked at 945; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09GG: sources/stacks-fields/pinned-original-source.tex, source lines 878–878. Prerequisite pointer invoked at 948; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let E/k be a field extension. Then the elements of E algebraic over k form a subextension of E/k.*

Proof. Let $\alpha, \beta \in E$ be algebraic over k. Then $k(\alpha, \beta)/k$ is a finite extension by Lemma 09GH. It follows that $k(\alpha+\beta) \subset k(\alpha, \beta)$ is a finite extension, which implies that $\alpha + \beta$ is algebraic by Lemma 09GG. Similarly for the difference, product and quotient of α and β. □

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